# JAVA PROGRAM REPORT

## Single Inheritance in Java

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|---|---|
| **Programming Language:** | Java |
| **Topic:** | Single Inheritance |
| **Program File:** | SingleInheritance.java |

## Objective

To demonstrate single inheritance in Java, where the Car class inherits the start() method from the Vehicle class and defines its own drive() method.

# Java Source Code

```
class Vehicle {
    void start() {
        System.out.println("Vehicle starts");
    }
}

class Car extends Vehicle {
    void drive() {
        System.out.println("Car is driving");
    }
}

public class SingleInheritance {
    public static void main(String[] args) {

        Car c = new Car();

        c.start();
        c.drive();
    }
}
```

# Program Explanation

**1. Vehicle Class:** The parent class defines start(), which prints “Vehicle starts”.

**2. Car Class:** Car extends Vehicle, so it inherits start() and adds the drive() method.

**3. Single Inheritance:** One child class, Car, inherits from one parent class, Vehicle.

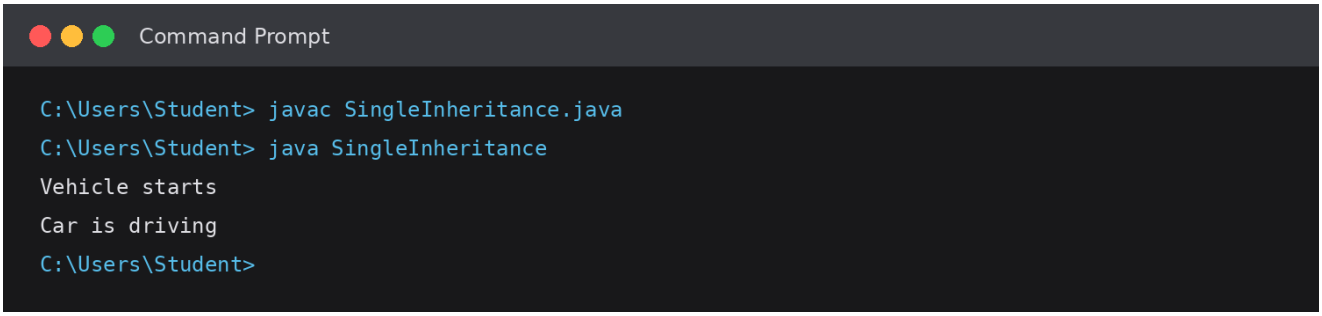
**4. Main Method:** The program creates a Car object and calls the inherited start() method followed by the drive() method.

## Expected Output

```
Vehicle starts
Car is driving
```

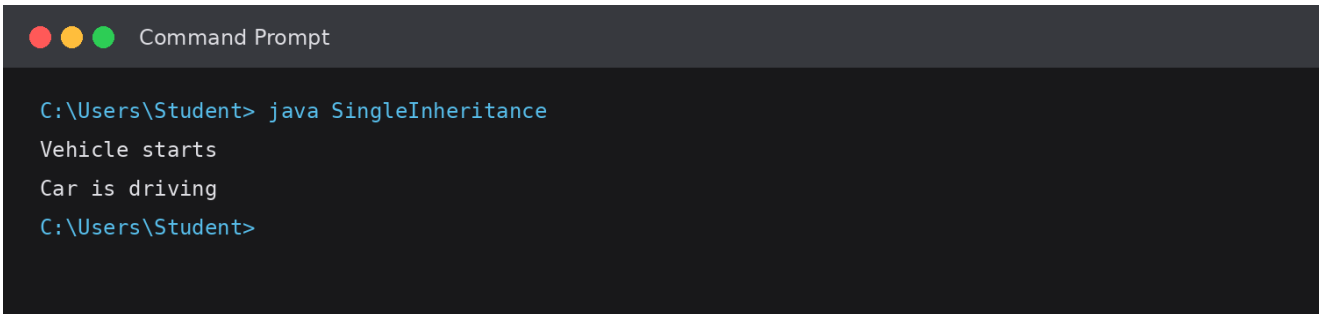
# Execution and Output Screenshots

## Test Case 1: Compilation and Execution




**Verification:** The program compiles and displays the inherited Vehicle message and the Car-specific message.

## Test Case 2: Re-execution Verification




**Verification:** Re-running the class produces the same output sequence.

# Test Case Summary

| Test Case | Expected Result | Actual Result | Status |
|---|---|---|---|
| Test Case 1 | Both methods execute in order. | Output matches expected messages. | **PASS** |
| Test Case 2 | Same output on re-execution. | Output remains consistent. | **PASS** |

# Conclusion

The program demonstrates single inheritance in Java. Car inherits the start() method from Vehicle and adds its own drive() behavior. The output confirms that the inherited and class-specific methods execute as expected.